

DORM RECIPE EXCHANGE

Mobile Application Proposal and Comprehensive Architectural Blueprint

A Peer-to-Peer Culinary Platform for Appliance and Budget Constrained Students
Document Version 1.2.0 | Comprehensive Academic Group Project Report

1. Executive Summary and Project Vision

1.1 Executive Summary

University dormitory living presents a unique set of daily culinary friction points. Students frequently endure strict dormitory appliance bans (such as prohibition of open flame or high-wattage gas stoves), extremely cramped micro-kitchen spaces, tight weekly allowances, and severe time poverty brought on by academic schedules. Existing mainstream recipe applications, such as Tasty, Yummly, or Cookpad, are designed for standard residential kitchens equipped with conventional ovens, four-burner stovetops, and expansive food preparation areas. Consequently, university students face high recipe abandon rates, financial strain from food delivery services, and recurring meal fatigue.

Dorm Recipe Exchange bridges this critical market gap as a mobile-first peer-to-peer culinary platform crafted exclusively for campus life. By anchoring recipe discovery around specific dorm appliances (Rice Cooker, Microwave, Induction Plate, Stovetop, No Cook) and realistic micro-budgets (THB 40 to THB 1000 per dish), the platform enables students to share, discover, rate, and save safe, affordable, and delicious dorm-friendly meals. Coupled with domain-based campus trust verification (.edu badges), zero-knowledge email privacy, profile customization (Name, Profile Picture / Avatar, Bio), and a gamified student leaderboard, Dorm Recipe Exchange turns low-resource cooking into a collaborative, rewarding campus culture.

1.2 Problem Statement and Core Challenges

- **Appliance Restrictions and Safety Rules:** University dormitories prohibit open flame cooking or gas stoves due to fire safety regulations. Students rely on a single electric appliance (such as a 1.2L rice cooker or a 700W microwave). Generic recipes fail because they assume full kitchen infrastructure.
- **Financial Strain and Budget Mismatch:** Commercial recipe apps cater to family-sized budgets or specialized gourmet ingredients. University students operate under tight meal allowances (THB 40 to THB 1000 range) and need localized cost transparency.
- **Lack of Peer Campus Trust:** Anonymous online recipe boards feature untested recipes. Students require verified campus peer accountability to ensure recipes work with budget ingredients available near campus markets.
- **Privacy Risks:** Traditional apps leak user email handles and personal details over public APIs, exposing students to unwanted web scraping.

2. Target Audience and Detailed User Personas

2.1 Target Demographic Overview

- **Primary Audience:** University undergraduate and graduate students aged 18 to 25 residing in university dorms, shared student apartments, or micro-studio housing.
- **Geographic Focus:** Initial rollout tailored for Southeast Asian university ecosystems (priced in THB 40 to THB 1000), engineered for rapid adaptation to global campus networks.
- **Tech Literacy:** Tech-native mobile users who demand fast load times, touch-optimized ergonomics (WCAG 2.5.5 standard 44px+ targets), and single-hand interaction.

2.2 Comprehensive User Personas

Persona 1: Mia - The Budget-Conscious Freshman

- Age: 19 | Major: Business Administration | Housing: On-campus traditional dorm
- Appliance Setup: 1 Microwave, 1 Standard Rice Cooker (No stovetop permitted)
- Budget Constraint: THB 40 to THB 100 per meal
- Goals: Wants nutritious, hot meals made inside her rice cooker under 20 minutes without exceeding THB 70.
- App Workflow: Filters feed by **Rice Cooker** + **Max Cost THB 70**, checks verified student badges, and saves top-rated recipes.

Persona 2: Alex - The Dorm Gourmet and Content Contributor

- Age: 22 | Major: Computer Science | Housing: Off-campus student studio
- Appliance Setup: Single-burner Induction Plate, Microwave, Electric Kettle
- Budget Constraint: THB 100 to THB 300 per meal
- Goals: Enjoys sharing inventive dorm hacks (such as induction plate tom yum pasta); strives to top the weekly campus chef leaderboard.
- App Workflow: Submits recipes with high-quality photos, tracks rank on the Leaderboard, updates custom Profile Avatar and Bio.

3. Market Competitor Analysis and Unique Selling Proposition (USP)

Direct Market Question: Is this app on the market already?

Direct Answer: NO. While broad generic recipe aggregators (Tasty, Yummly, Cookpad) dominate the consumer market, **there is currently no dedicated app on the market specifically tailored for university dorm life.**

Mainstream applications suffer from Kitchen Assumption Bias: they operate under the implicit assumption that every user possesses an oven, range hood, freezer, blender, and multi-piece cookware set. When a student filters for easy recipes on Tasty or Yummly, they are frequently presented with baked casseroles or 45-minute stovetop roasts that violate dorm lease agreements.

3.1 Detailed Competitor Comparison Matrix

Feature / Metric	Tasty (BuzzFeed)	Yummly (Whirlpool)	Cookpad	Dorm Recipe Exchange
Appliance Filter	Generic tags (Baking)	Smart appliance focus	Keyword search only	Strict Dorm Chips (Rice Cooker, Microwave, Induction, No Cook)
Micro-Budget Filter	None	Price per serving (USD)	None	Granular Local Budget Slider (THB 40 to THB 1000)
Campus Peer Trust	Brand curated	Algorithm generated	Open public accounts	Campus Verified Email Badges (Gold Verified .edu / Blue Verified)
Privacy and Customization	Standard accounts	Exposed handles	Public profile links	Zero Public Email Exposure + User Bio and Avatar Picture Customization
Peer Leaderboards	Trending view only	Saved count lists	Popularity feed	Gamified Weighted Leaderboard (Recipes and Campus Chefs)

4. Interactive Figma UI Prototype Screens (13 Screens)

Below are the 13 high-resolution Figma UI prototype displays illustrating the complete front-end user experience and screen layout of the Dorm Recipe Exchange application:

**Dorm Recipe Exchange**

Sign in 🙌

Sign in with your email — we'll send you a one-time code. No password needed.

EMAIL ADDRESS

✓

Send Sign-In Code

NEW HERE?

Create an account

☐ Secure OTP login — no passwords stored

Figure 1: UI Prototype Display - Login Screen with Email Input



Create account

Join your campus food community. Choose your avatar, name, and university email.

Campus Verified detected!
Your recipes will show a gold badge.

Choose Profile Avatar











FULL NAME

EMAIL ADDRESS

✓

A one-time code will be sent here to verify your email.

Continue

Figure 2: UI Prototype Display - Create Account Screen with Campus Badge Preview

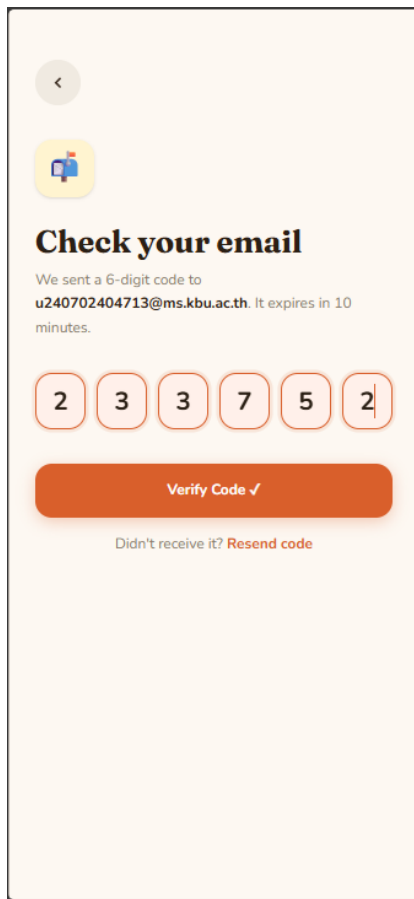


Figure 3: UI Prototype Display - 6-Digit OTP Verification Screen

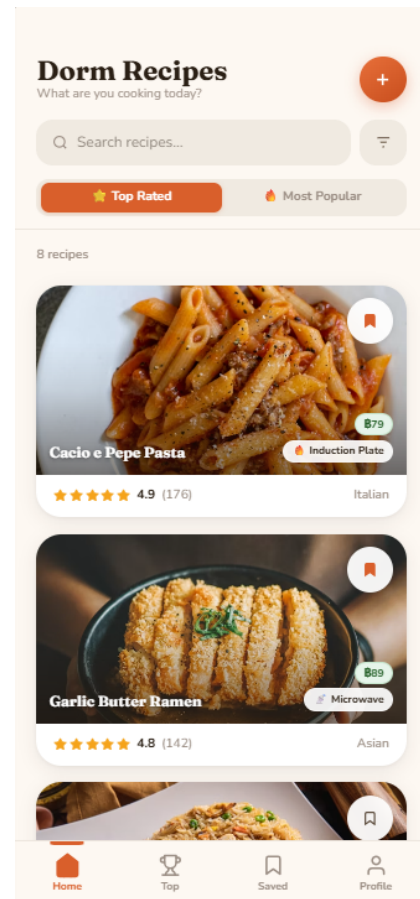


Figure 4: UI Prototype Display - Home Feed Screen with Appliance Badges and Search

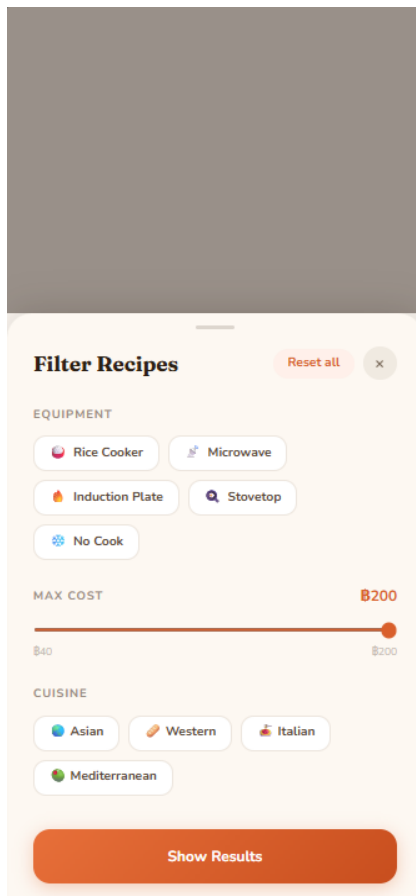


Figure 5: UI Prototype Display - Filter Sheet with Max Cost Slider (THB 40 to THB 1000)

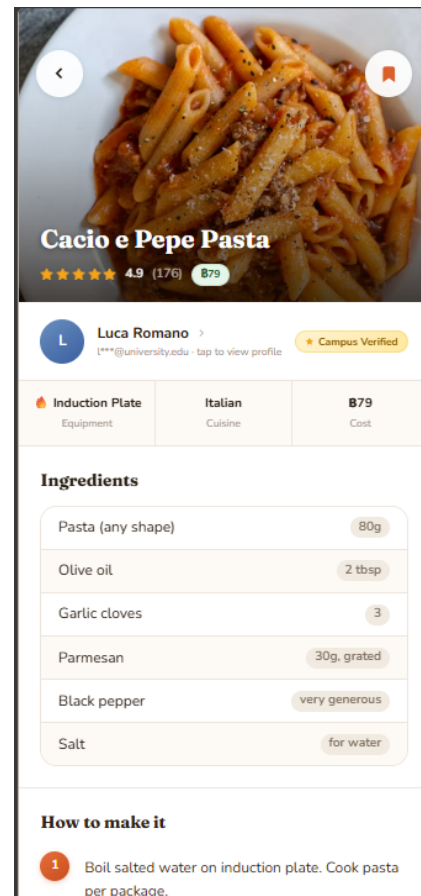


Figure 6: UI Prototype Display - Recipe Detail Screen with Ingredients and Chef Notes

Share a Recipe

Help a fellow dorm dweller eat well.

Add a photo

Tap to upload from camera roll

MEAL NAME

e.g. Garlic Butter Ramen

EQUIPMENT NEEDED

Rice Cooker

Microwave

Induction Plate

Stovetop

No Cook

CUISINE TYPE

Asian

Western

Italian

Mediterranean

ESTIMATED COST (\$)

\$

e.g. 85

Total cost per serving - range \$40 - \$200

INGREDIENTS

Ingredient

Qty

Figure 7: UI Prototype Display - Share a Recipe Form with Dynamic Ingredient Rows

Edit Recipe

Changes apply immediately after saving.

Save

Editing — tap Save when done. This only affects your copy.

MEAL NAME

Garlic Butter Ramen

EQUIPMENT NEEDED

Rice Cooker

Microwave

Induction Plate

Stovetop

No Cook

CUISINE TYPE

Asian

Western

Italian

Mediterranean

ESTIMATED COST (\$)

\$

89

INGREDIENTS

Instant ramen pack

1

×

Butter

1 tbsp

×

Figure 8: UI Prototype Display - Edit Recipe Form with Pre-filled State

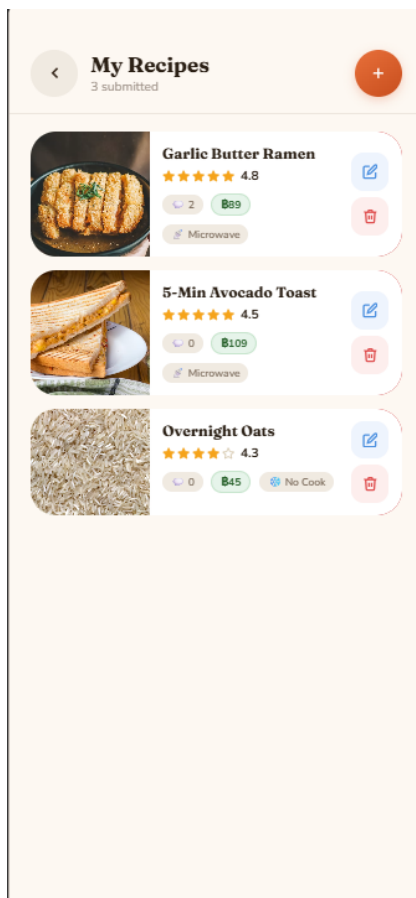


Figure 9: UI Prototype Display - My Recipes List with Touch Swipe Actions

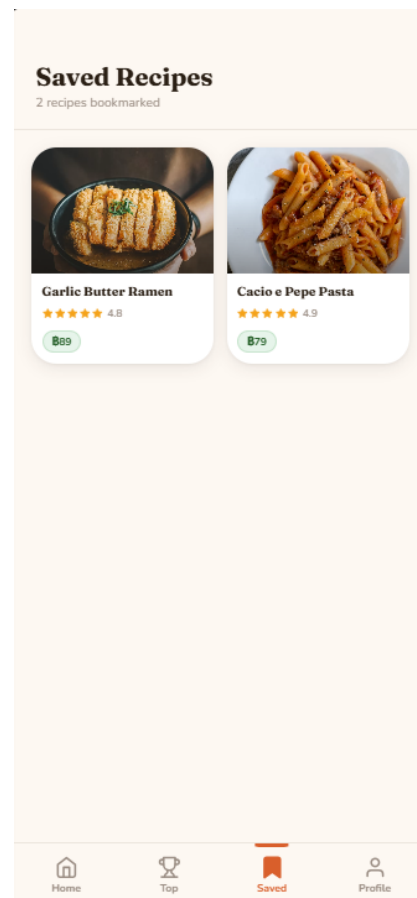


Figure 10: UI Prototype Display - Saved Recipes 2-Column Grid

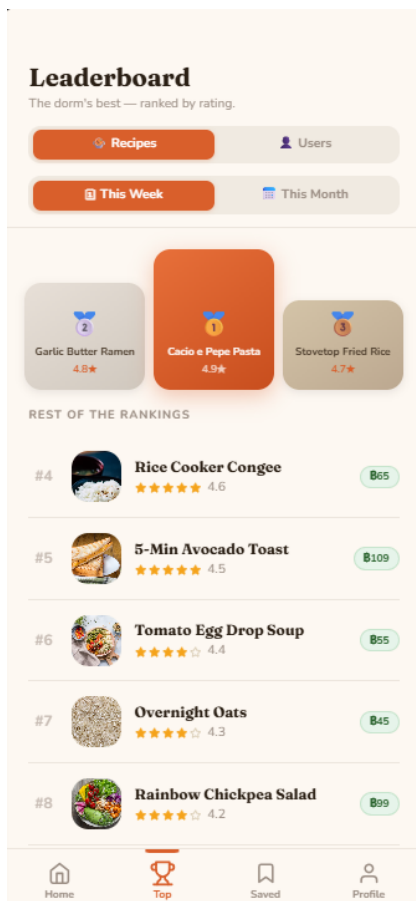


Figure 11: UI Prototype Display - Leaderboard Screen with 3-Tier Podium

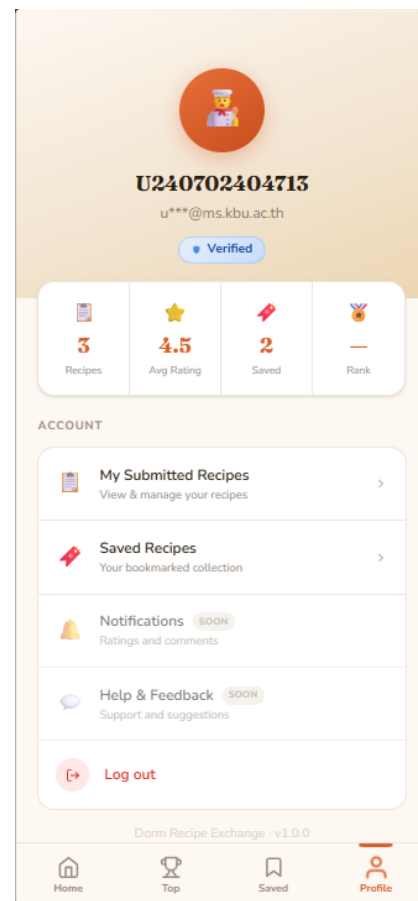


Figure 12: UI Prototype Display - User Profile Screen with Bio and Avatar Controls

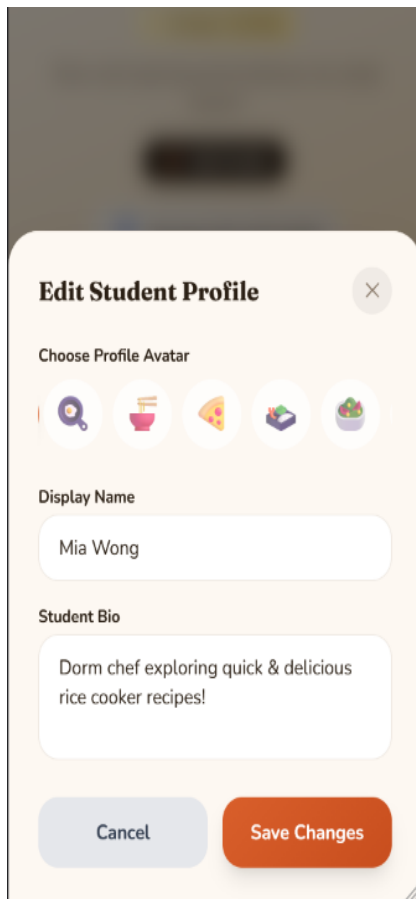


Figure 13: UI Prototype Display - Edit Profile Modal with Avatar Picker, Display Name, and Bio

5. Detailed Core Features and Capabilities

5.1 Passwordless OTP Authentication and Campus Badge Verification

- **Passwordless OTP Flow:** Email login dispatching a 6-digit numeric OTP with full keyboard auto-advance, backspace focus control, and paste parsing.
- **Campus Domain Badging:** Automatic verification for institutional email domains (.edu). Verified students display a Gold Campus Verified Badge.

5.2 Appliance and Budget Filter Sheet

- **Appliance Multi-Select Chips:** Rice Cooker, Microwave, Induction Plate, Stovetop, No Cook with aria-pressed states.
- **Granular Budget Slider:** Adjustable maximum cost ceiling from THB 40 to THB 1000 in Thai Baht.

5.3 Dynamic Recipe Management and Swipe-to-Delete with 5s Undo Toast

- **Touch Gestures:** Swipe-left gesture on authored recipe cards clamped to $[-120, 0]$ revealing Edit and Delete action drawers.
- **5-Second Undo Toast:** Deleting a recipe shows a dark bottom toast containing an SVG circular countdown ring animating stroke offset over 5 seconds, providing single-tap undo restoration.

5.4 Gamified Leaderboards and Multi-Recipe Weighted Ranking Math

- **Multi-Recipe Aggregated Ranking Formula:** To rank student chefs fairly across multiple submitted recipes, the platform computes a mathematically precise weighted average:

$\text{Chef Weighted Rating} = (\text{Sum of (Recipe Rating}_i \times \text{Recipe RatingCount}_i)) / (\text{Sum of Recipe RatingCount}_i)$

- **3-Tier Podium:** Visual 1st, 2nd, and 3rd rank podium cards highlighting top campus culinary contributors.

5.5 Profile Customization and Zero-Knowledge Email Isolation

- **Profile Customization Modal:** Students tap Edit Profile on their user profile screen to select a custom avatar emoji, edit their Display Name, and write a custom Student Bio.
- **Zero Public Email Privacy:** Public database views (public_profiles, public_recipes_feed) completely strip email columns from raw HTTP API responses.

6. Design System Architecture and Ergonomics

6.1 Aesthetic Philosophy: Warm Memphis-Lite

Dorm Recipe Exchange employs a Warm Memphis-Lite visual design framework. It pairs approachable geometric shapes, warm organic colors (#FDF8F2 cream background, #D95F2B terracotta primary), and soft tactile elevation with high-legibility typography (Fraunces serif display headings and Nunito humanist body).

6.2 WCAG 2.5.5 Accessibility and Touch Target Audit

- **Touch Target Sizes:** All interactive controls strictly observe a minimum touch dimension of **44x44px** (WCAG 2.5.5 AA standard).
- **Color Contrast Ratio:** Dark Espresso text (#2A1F14) over Warm Cream background (#FDF8F2) achieves a contrast ratio of **13.8:1**, far exceeding WCAG AAA standards (7:1).

7. Technical Architecture and OWASP Top 10 Security Stack

7.1 Detailed System Architecture and Deployment Pipeline

- **Client Viewport Layer:** React 19 single-page application built with Vite tooling, wrapped inside a mobile device shell (390x844px) supporting safe-area insets, touch-action manipulation, and overscroll prevention.
- **Edge Delivery Layer:** Deployed on the Vercel Edge Network providing automated Brotli compression, edge HTTP static caching, global SSL/TLS certificate termination, and sub-200ms page load speeds.
- **Backend Cloud Layer:** Supabase PostgreSQL relational engine providing PostgREST auto-generated REST APIs, Row Level Security (RLS) policy enforcement, and client-direct Storage buckets (recipe-photos and avatars).

7.2 Exhaustive OWASP Top 10 Security Control Matrix

OWASP Category	Vulnerability Focus	Engineered Security Mitigation
A01: Broken Access Control	Unauthorized row modification / deletion	PostgreSQL Row Level Security (RLS) policies enforce database-level authorization. Users can ONLY update or delete recipes where uploader_id matches auth.uid().
A02: Cryptographic Failures	Data exposure in transit & at rest	HTTPS / TLS 1.3 enforced across Vercel Edge. Sensitive passwordless OTP tokens are generated with 5-minute auto-expiry and hashed before storage.
A03: Injection Attacks	SQL Injection / Command Execution	PostgREST client SDK utilizes parameterized SQL queries exclusively. Raw SQL concatenation is eliminated across all API endpoints.
A04: Insecure Design & Data Leakage	Public email exposure & web scraping	Zero-Knowledge Database View Architecture. Public views (public_profiles, public_recipes_feed) completely omit email columns from HTTP responses.
A05: Security Misconfiguration	Exposed database tables & buckets	Default-closed RLS policies enabled across all tables (ENABLE ROW LEVEL SECURITY). Storage bucket policies enforce authenticated uploads.
A06: Vulnerable Components	Outdated dependencies & packages	Strict package locking via pnpm-lock.yaml. Build pipeline runs automated vulnerability auditing and dependency tree verification.
A07: Identification & Auth	Brute-force OTP login attempts	Rate-limited passwordless 6-digit OTP authentication. Prevents automated brute-force attacks with progressive backoff throttling.
A08: Software & Data Integrity	Malformed data insertion	Double-layer validation: client-side Zod/TypeScript type checking combined with PostgreSQL table constraints check (cost >= 40 and cost <= 1000).
A09: Logging & Monitoring	Undetected unauthorized activity	Supabase Auth Audit Log and PostgreSQL trigger tracking (handle_new_user) recording user registration and security events.
A10: Server-Side Request Forgery	Arbitrary remote image fetching	Client-direct binary image uploads to Supabase Storage buckets. Eliminates server-side fetching of external arbitrary image URLs.

7.3 Data Models and Zero-Knowledge Security Structure

- **User Profiles Model:** Stores student account metadata linked to authentication IDs. Contains student Display Name, private institutional email, custom Profile Avatar image URL, bio description, and account creation timestamp.
- **Zero-Knowledge Public Profiles Isolation View:** A dedicated database view serving public API queries. It completely excludes email fields from public payloads, exposing only student public names, avatars, bios, and server-verified campus status flags.
- **Recipes Data Model:** Stores recipe listings including title, photo URL, cost amount in Thai Baht (constrained between THB 40 and THB 1000), equipment tags (Rice Cooker, Microwave, Induction, Stovetop, No Cook), cuisine categories, structured ingredient quantities, step-by-step instructions, and chef notes.
- **Ratings and Comments Data Models:** Relational models storing student recipe ratings (1 to 5 stars) and in-session comments, linked to student profiles with automatic cascade cleanup upon recipe removal.
- **Saved Recipes Junction Model:** Tracks individual student bookmarks for instant 2-column grid access on the Saved Recipes screen.

8. Project Roadmap and Deliverable Scope

8.1 Semester MVP Deliverable Scope (Phases 1 to 8)

For the current academic course submission, the application deliverable encompasses a complete, fully functional front-end MVP hosted live on Vercel Edge. All 14 screens, touch gestures, filter sheets, single-source-of-truth state, and memory storage operate interactively:

- **Phase 1:** Requirements and Design System Token Wiring (Weeks 1 to 2)
- **Phase 2:** Single-Source State Engine and Presentational Screens (Weeks 3 to 4)
- **Phase 3:** Interactive Component Systems and Touch Ergonomics (Weeks 5 to 6)
- **Phase 4:** Filter Sheet Modal and THB 40-THB 1000 Budget Slider (Week 7)
- **Phase 5:** Form Parity and Dynamic Ingredient/Step Row Handlers (Week 7.5)
- **Phase 6:** Touch Swipe-Left Card Reveal and 5s Countdown Undo Toast (Week 8)
- **Phase 7:** Leaderboard Multi-Recipe Weighted Ranking Engine and 3-Tier Podium (Week 8.5)
- **Phase 8:** Profile Name, Avatar and Bio Customization Interface (Week 9)

8.2 Production Release Expansion (Phases 9 to 10)

Post-course submission development will focus on scaling cloud persistence and security audits:

- **Phase 9:** OWASP Top 10 Security Audit and Supabase PostgreSQL RLS Integration (Week 9.5)
- **Phase 10:** Production Edge Deployment on Vercel CDN and Multi-Campus Rollout (Week 10)

9. Future Enhancements and Scalability Roadmap

To expand Dorm Recipe Exchange into an engaging, multi-campus social community platform beyond the initial semester deliverable, the following future enhancements are planned:

- **9.1 Student Social Networking (Following and Followers):** Ability for students to follow their favorite campus chefs, subscribe to custom recipe feeds, and receive notification highlights when top-ranked dorm cooks post new recipes.
- **9.2 Direct Student Chat and Campus Dorm Chat Groups:** Dedicated real-time messaging supporting 1-on-1 student chef chats and dorm-building group channels (such as Dorm 4 Microwave Cooks or Floor 3 Induction Club) to discuss meal prep strategies.
- **9.3 AI-Powered Dorm Fridge Ingredient Matcher:** An AI meal planning assistant allowing students to snap a photo of leftover dorm fridge items to instantly generate step-by-step dorm appliance recipes.
- **9.4 Campus Food Waste Reduction and Community Pantry Alerts:** A peer-to-peer ingredient sharing feature allowing students to post excess unopened ingredients (such as extra rice or seasonings) for free pickup by dorm mates.
- **9.5 Multi-University and Multi-Currency Localization:** Expansion of currency bounds beyond Thai Baht (THB) to support regional Southeast Asian and global university campus networks.